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Nemotron-4 15B Technical Report

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Abstract

We introduce Nemotron-4 15B, a 15-billion-parameter large multilingual language model trained on 8 trillion text tokens. Nemotron-4 15B demonstrates strong performance when assessed on English, multilingual, and coding tasks: it outperforms all existing similarly-sized open models on 4 out of 7 downstream evaluation areas and achieves competitive performance to the leading open models in the remaining ones. Specifically, Nemotron-4 15B exhibits the best multilingual capabilities of all similarly-sized models, even outperforming models over four times larger and those explicitly specialized for multilingual tasks.

1 Introduction

Recently published efforts (Hoffmann et al., 2022; Touvron et al., 2023a,b; Yang et al., 2023; Jiang et al., 2023) in language model pre-training have been inspired by Chinchilla scaling laws (Hoffmann et al., 2022), which argue for scaling data along with model size given a fixed compute budget, compared to past work that only scaled the size of the model (Kaplan et al., 2020; Brown et al., 2020; Smith et al., 2022; Rae et al., 2022; Scao et al., 2023). For example, (Hoffmann et al., 2022) shows that given two roughly IsoFLOP GPT models with a similar data distribution, a 65-billion-parameter model on 1.4 trillion tokens and a 280-billion-parameter model on 300 billion tokens, the 65B model has better accuracy on downstream tasks.

This trade-off of allocating compute towards training on more data as opposed to increasing model size is particularly appealing from an inference perspective, reducing latency and the amount of compute needed to serve models. As a consequence, a major focus of language modeling training efforts has shifted to collecting high-quality multi-trillion token datasets from public sources such as Common Crawl. We continue this trend by introducing Nemotron-4 15B which was trained on 8 trillion tokens of English, multilingual,

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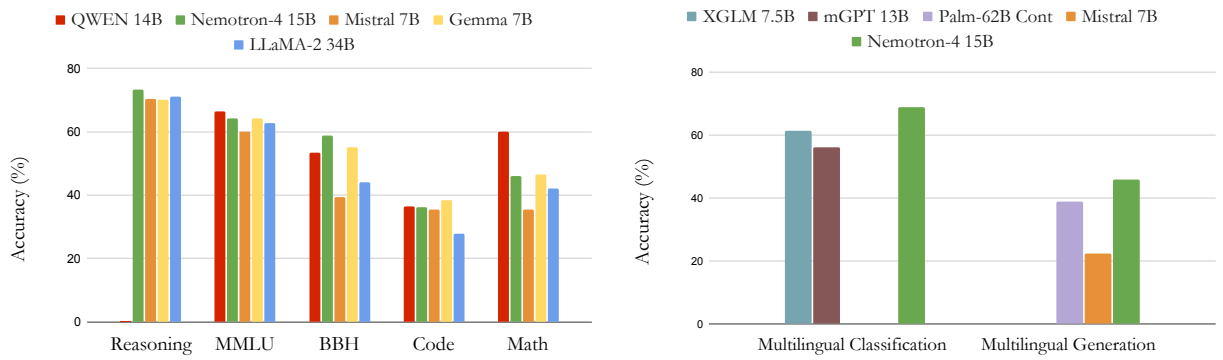

Figure 1: Comparison of Nemotron-4 15B across seven evaluation areas against similarly sized models. The composition of tasks that form each evaluation area can be found, along with more detailed evaluation results, in Section 3

and coding text and was developed to be the best general-purpose large language model (LLM) that can fit on a single NVIDIA A100 or H100 GPU.

As demonstrated in Figure 1, Nemotron-4 15B exhibits high downstream accuracies across a wide range of English, code, and multilingual evaluation areas. In comparison to leading similarly-sized, open models we show that Nemotron-4 15B is significantly better than LLaMA-2 34B (Touvron et al., 2023b), which has over twice the number of parameters, and is better than Mistral 7B (Jiang et al., 2023) on all English evaluation areas. Additionally, Nemotron-4 15B achieves competitive accuracies to QWEN 14B (Bai et al., 2023) and Gemma 7B (Gemma Team, 2024). In a comparison across a wide range of programming languages, we find that Nemotron-4 15B achieves better average accuracy, and in particular on low-resource programming languages, than Starcoder (Li et al., 2023), a code-specific model, and Mistral 7B. As Nemotron-4 15B was trained on significant amount of multilingual data, it is currently the state-of-the-art general purpose model in its size class on all multilingual benchmarks. We find that Nemotron-4 is better than PALM 62B-Cont (Slav Petrov and et al., 2023), and also outperforms multilingual-specific models such as XGLM (Lin et al., 2022) and mGPT (Shliazhko et al., 2022).

Number of transformer layers	Hidden dimension	Number of attention heads	Number of KV heads	Sequence length	Vocabulary size
32	6144	48	8	4096	256,000

Table 1: Key hyper-parameters affecting size of Nemotron-4 15B.

2 Architecture Details

Nemotron-4 uses a standard decoder-only Transformer architecture (Vaswani et al., 2017), with causal attention masks. Exact hyper-parameters affecting size are shown in Table 1. Nemotron-4 has 3.2 billion embedding parameters and 12.5 billion non-embedding parameters. We use Rotary Position Embeddings (RoPE) (Su et al., 2021), SentencePiece tokenizer (Kudo and Richardson, 2018), squared ReLU activations in the MLP layers, no bias terms, dropout rate of zero, and untied input-output embeddings. We use grouped

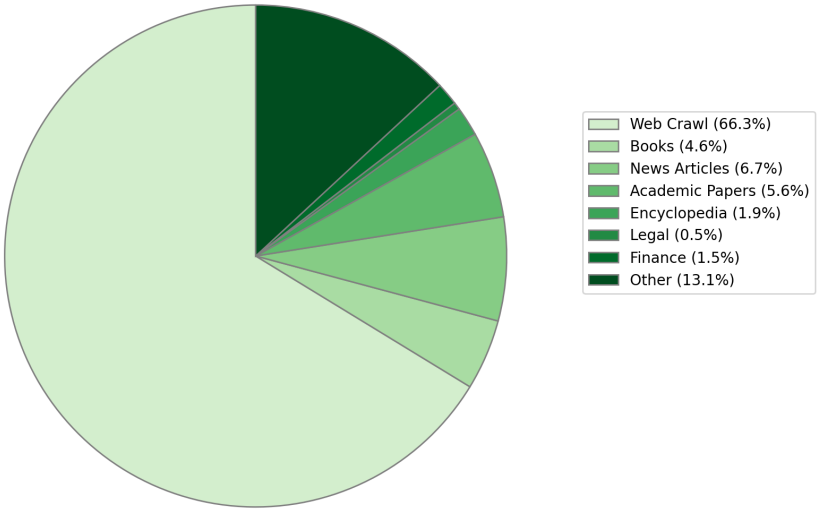

Figure 2: Data composition of the English tokens used for pre-training

query attention (GQA) (Ainslie et al., 2023) for faster inference and lower memory footprint.

Data. We train Nemotron-4 15B on a pre-training dataset consisting of 8 trillion tokens. At a high-level, the data blend is split into three different types of data: English natural language data (70%), multilingual natural language data (15%), and source-code data (15%).

The English corpus consists of curated documents from a variety of sources and domains including web documents, news articles, scientific papers, books, etc and the distribution used in our pre-training set is highlighted in Figure 2. The code and multilingual data consists of a diverse set of natural and programming languages. We find that appropriately sampling tokens from these languages is key to strong accuracies in these domains. We share the distributions used for both code and multilingual tokens in our pre-training dataset in Figure 3 and Figure 4 respectively.

In constructing the pre-training corpus, we remove any possible duplicates via document-level exact and near-deduplication (Jennings et al., 2023). We additionally applied document-level quality filtering across our corpus using a language-model based filtering approach similar to (Wenzek et al., 2019) in addition to a series of heuristic filters as described in (Rae et al., 2022) and (Raffel et al., 2020).

We train a BPE tokenizer in SentencePiece (Kudo and Richardson, 2018) on data that is randomly sampled from the final 8T token dataset. To have better coverage of low-resource languages in the tokenizer, we upsample non-English data relative to the final training dataset distribution. Our tokenizer preserves whitespaces (including leading and trailing ones), splits numbers into their individual digits (Chowdhery et al., 2022), and relies on byte-level backoff to handle unknown character sequences. The final vocabulary size is 256,000 tokens.

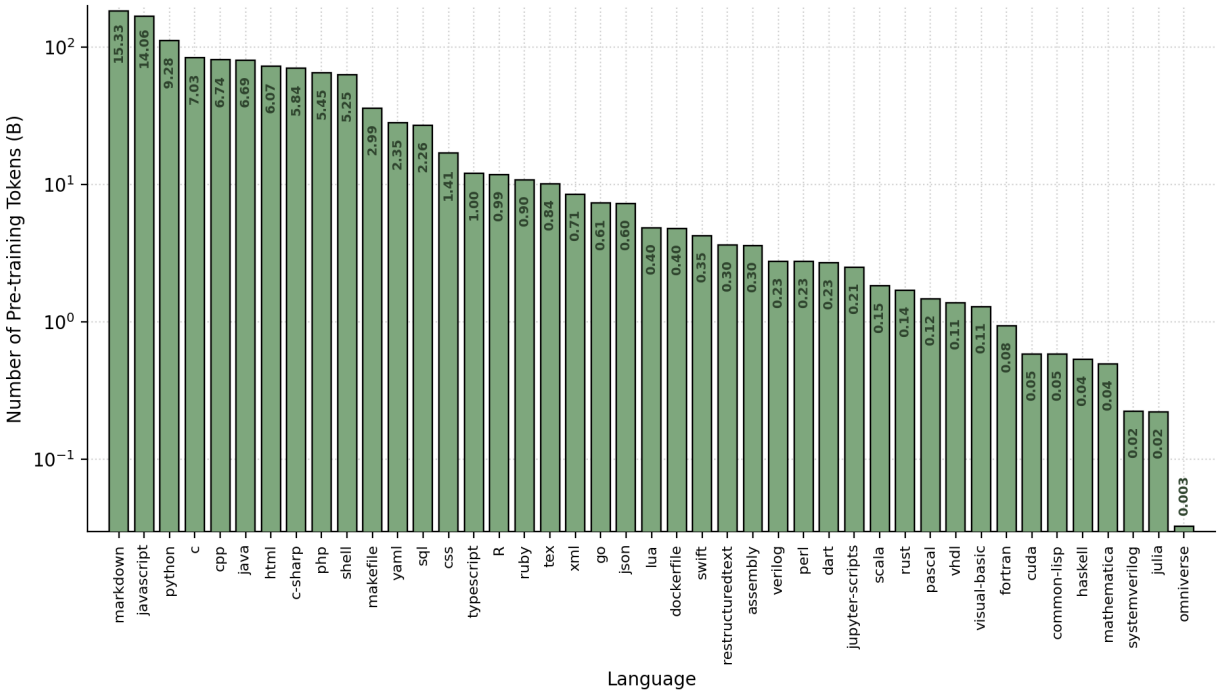

Figure 3: Data distribution of the 43 programming languages used for pre-training. The number within each bar indicates the percent of the overall code distribution that an individual language comprises.

Pre-training. Nemotron-4 was trained using 384 DGX H100 nodes; each node contains 8 H100 80GB SXM5 GPUs based on the NVIDIA Hopper architecture (NVIDIA, 2022). Each H100 GPU has a peak throughput of 989 teraFLOP/s when doing 16-bit floating point (`bfloat16`) arithmetic without sparsity. Within each node, GPUs are connected by NVLink and NVSwitch (nvl); the GPU-to-GPU bandwidth is 900 GB/s (450 GB/s in each direction). Each node has 8 NVIDIA Mellanox 400 Gbps HDR InfiniBand Host Channel Adapters (HCAs) for inter-node communication.

We used a combination of 8-way tensor parallelism (Shoeybi et al., 2019) and data parallelism to train the model; we also use a distributed optimizer to shard the optimizer state over the data-parallel replicas. The degree of data parallelism was varied from 96 to 384 as the batch size was ramped up. Table 2 summarizes the 3 stages of batch size ramp, and includes the per-iteration time and model FLOP/s utilization (MFU) (Chowdhery et al., 2022; Korthikanti et al., 2022). MFU quantifies how efficiently the GPUs are utilized in model training. Training was completed in approximately 13 calendar days.

Data-parallel size	GPUs	Iteration time (secs)	MFU (%)	Batch size	Tokens (B)	Time (days)
96	768	0.57	34.3	384	200	0.8
192	1,536	0.58	33.3	768	200	0.4
288	2,304	0.64	30.5	1,152	7,600	11.9

Table 2: Batch size rampup schedule, along with time and efficiency metrics for the Nemotron-4 15B parameter model.

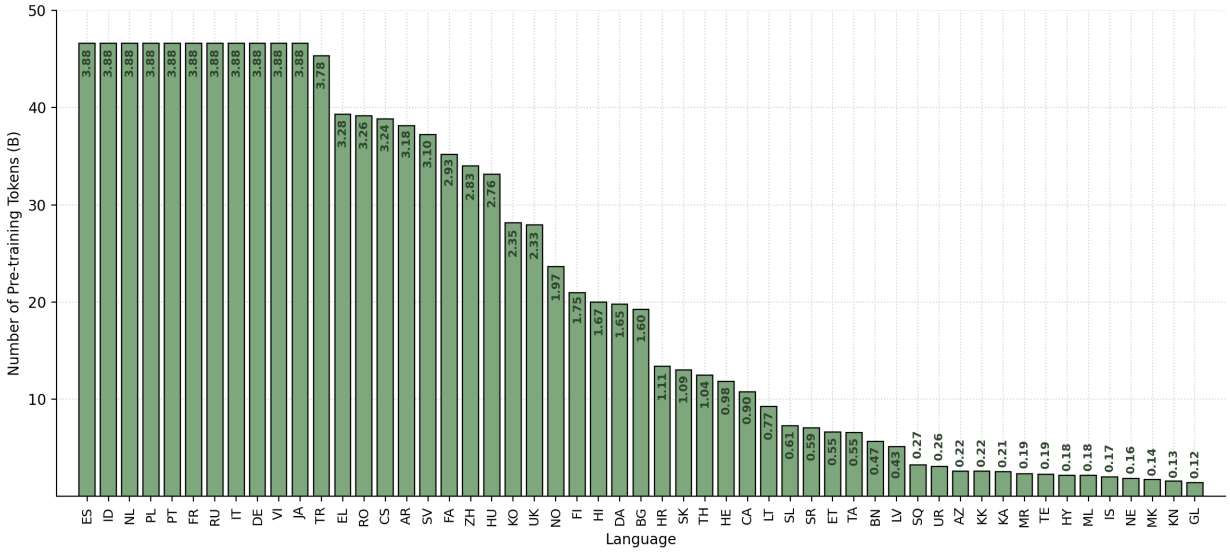

Figure 4: Data distribution of the 53 natural languages, aside from English,we used for pre-training. The number within each bar indicates the percent of the overall multilingual distribution that an individual language comprises.

Continued Training. Similar to recent work (Google, 2023), we find that switching the data distribution and learning rate decay schedule at the end of model training greatly improves model quality. Concretely, after having trained over the entirety of our 8T pre-training dataset, we use the same loss objective and perform continued training on small number of tokens in comparison to the pre-training tokens.

In this additional phase of continued training, we utilize two distinct data distributions. The first distribution is where the majority of tokens during continued training are sampled from. It utilizes tokens that have already been introduced during pre-training but with a distribution that places larger sampling weight on higher quality sources. The second distribution introduces a small number of benchmark-style alignment examples to better allow the model to respond to such questions in downstream evaluations while also up-weighting data sources that come from areas of low model performance. In accompaniment with a learning rate schedule that prioritizes a steeper slope of decay than magnitude of learning rate, we find that such an ordering and style of data distributions allows for the model to gently transition from the pre-training dataset and better learn newly emphasized data areas.

3 Results

We evaluate Nemotron-4 15B on a variety of downstream evaluation areas covering a diverse range of tasks and domains. In all evaluations, we adhere to the standardized task setup and share the exact settings used. The covered evaluation categories include:

- **Commonsense Reasoning (0-shot):** SIQA (Sap et al., 2019), ARC easy and challenge (Clark et al., 2018), PIQA (Bisk et al., 2020), Winogrande (Sakaguchi et al., 2020), and Hellaswag (Zellers et al., 2019)

- **Popular Aggregated Benchmarks:** MMLU (5-shot) (Hendrycks et al., 2020) and BBH (3-shot) (Suzgun et al., 2022)
- **Math:** GSM8K (8-shot with maj@1) (Cobbe et al., 2021)
- **Code:** Pass@1 scores on HumanEval (0-shot) (Chen et al., 2021), MBPP (3-shot) (Austin et al., 2021), and MultiPL-E (0-shot) (Cassano et al., 2023a)
- **Multilingual:** classification via XCOPA (0 and 4-shot) (Ponti et al., 2020), machine translation with FLORES-101 (8-shot) (Goyal et al., 2021), and generation tasks such as MGSM (8-shot) (Shi et al., 2022) and TyDiQA (1-shot) (Clark et al., 2020)

In our evaluations, we compare against a number of external decoder-only transformer language models and unless otherwise stated we use the numbers published in the reports of the corresponding models. For English and code tasks, we share detailed results for Nemotron-4 15B, LlaMA-2 13B and 34B (Touvron et al., 2023b), Mistral 7B (Jiang et al., 2023), Baichuan-2 13B (Yang et al., 2023), QWEN 14B (Bai et al., 2023), and Gemma 7B (Gemma Team, 2024). For multilingual benchmarks, we report results against PaLM 62B and 62B-cont (Chowdhery et al., 2022) as well as models specially trained for multilingual capabilities such as mGPT 13B (Shliazhko et al., 2022) and XGLM 7.5B (Lin et al., 2022).

3.1 Commonsense Reasoning

We use the LM-Evaluation Harness (Gao et al., 2021) to evaluate Nemotron-4 15B across all aforementioned tasks. Table 3 showcases that Nemotron-4 15B achieves the strongest average performance on this diverse set of tasks.

	Size	SIQA	ARC-c	ARC-e	PIQA	Winogrande	Hellaswag	AVG
LLaMA-2	13B	50.3	49.4	77.3	79.8	72.8	80.7	68.4
	34B	50.9	54.5	79.4	81.9	76.7	83.3	71.1
Baichuan-2	13B	-	-	-	78.1	-	70.8	-
QWEN	14B	77.9	84.4	90.3	79.9	-	80.2	-
Mistral	7B	47.0*	55.5	80.0	83.0	75.3	81.3	70.4
Gemma	7B	51.8	53.2	81.5	81.2	72.3	81.2	70.2
Nemotron-4	15B	60.9	55.5	80.9	82.4	78.0	82.4	**73.4**

Table 3: Results on standard reasoning benchmarks in the zero-shot setting. We report the average across all tasks where possible for a fair comparison. The values marked with $*$ are read from Gemma Team (2024)

3.2 Popular Aggregated Benchmarks

The MMLU (Hendrycks et al., 2020) and Big Bench Hard (BBH) (Suzgun et al., 2022) benchmarks have been developed as a challenging assessment of language models' capabilities on a wide range of tasks and domains. As seen from Table 4, Nemotron-4 15B achieves the best score on BBH across existing models

at its scale by nearly 7%. Additionally, Nemotron-4 is significantly better than LLaMA-2 70B model on BBH benchmark where LLaMA-2 70B attains a score of 51.2 and Nemotron-4 is 58.7. Nemotron-4 15B additionally attains a highly competitive MMLU score and its per-category performance on MMLU can be found in Table 11.

	Size	BBH	MMLU
LLaMA-2	13B	39.4	54.8
	34B	44.1	62.6
Baichuan-2	13B	48.8	59.2
QWEN	14B	53.4	**66.3**
Mistral	7B	39.5	60.1
Gemma	7B	55.1	64.3
Nemotron-4	15B	**58.7**	64.2

Table 4: Nemotron-4 15B attains highly competitive performance on popular aggregate benchmarks. The BBH result for Mistral is read from the figure in (Jiang et al., 2023).

3.3 Math and Code

Recently, large language models have been shown to be effective at both mathematical reasoning and a variety of coding tasks (Allal et al., 2023; Chowdhery et al., 2022; Touvron et al., 2023a). Table 5 highlights the performance of Nemotron-4 15B on such tasks. Specifically, on mathematical reasoning we find that Nemotron-4 15B achieves strong performance as it attains a similar score to Gemma 7B, but lags behind models such as Baichuan-2 and QWEN. On code tasks, we see that Nemotron-4 performs on par with QWEN 14B while remaining slightly behind Gemma 7B. Across both types of tasks, Nemotron-4 15B is able to outperform Mistral 7B and LlaMA-2 13B/34B.

	Size	GSM8K	HumanEval	MBPP
LlaMA-2	13B	28.7	18.3	30.6
	34B	42.2	22.6	33.0
Baichuan-2	13B	52.8	17.1	30.2
QWEN	14B	**60.1**	32.2	40.8
Mistral	7B	35.4*	30.5	40.2*
Gemma	7B	46.4	**32.3**	**44.4**
Nemotron-4	15B	46.0	31.6	40.6

Table 5: Comparative results on math and code benchmarks. As Mistral 7B reports MBPP performance on a different eval split and uses a different evaluation setting for GSM8K , we use the corresponding numbers reported in (Gemma Team, 2024)

Nearly all similarly-sized open models determine their code abilities solely based on performance on Python

	Size	JavaScript	Julia	Java	Lua	C++	C-Sharp	PHP	Shell	TypeScript	R	Scala	AVG
Starcoder	15B	30.8	23.0	30.2	23.9	31.6	21.0	26.1	10.5	32.3	15.5	27.6	24.2
Mistral	7B	34.2	22.0	26.0	25.3	29.1	22.8	27.9	8.9	28.5	11.8	22.2	23.6
Nemotron-4	15B	28.6	24.8	24.8	24.2	35.4	21.1	27.3	8.9	32.9	18.6	27.3	**24.5**

Table 6: Nemotron-4 15B attains high competency in coding performance across a broad range of programming languages. Results for Mistral are from our runs of Mistral in the same setting as Nemotron-4.

related tasks – disregarding an evaluation of their capabilities on other programming languages. In Table 6, we demonstrate results of Nemotron-4 15B on the Multiple-E (Cassano et al., 2023b) benchmark across 11 diverse programming languages and compare it against Mistral 7B and Starcoder (Li et al., 2023), a 15B parameter model that has been specially trained for code. We find that Nemotron-4 15B attains strong coding performance across a wide assortment of programming languages and outperforms both Starcoder and Mistral 7B on average. We especially highlight the superior performance of Nemotron-4 15B on low-resource programming languages such as Scala, Julia, and R.

3.4 Multilingual

We demonstrate the outstanding multilingual ability of Nemotron-4 15B using four widely-studied benchmarks in previous works that cover a diverse range of high to low resource natural languages. For classification we use accuracy as the metric; for generative tasks, we use exact match; and for machine translation, we evaluate using the `sacreBLEU` (Post, 2018) implementation of `BLEU` (Papineni et al., 2002), using `spm-flores-101` tokenization to obtain spBLEU scores.

1. Classification: Cross-lingual Choice of Plausible Alternatives (XCOPA) (Ponti et al., 2020) tests causal commonsense reasoning in 11 languages

We compare Nemotron-4 15B to existing multilingual language models: XGLM (Lin et al., 2022) , mGPT (Shliazhko et al., 2022), and BLOOM (Scao et al., 2023). XGLM and mGPT are models specially trained to have improved multilingual ability by up-sampling the presence of non-English languages in the training data. In contrast, BLOOM, like Nemotron-4, is a general purpose language model that was trained on a combination of English, multilingual, and code data. In Table 7, we clearly see that Nemotron-4 achieves the best performance amongst all models – realizing almost a 12% improvement in the four-shot setting.

Mode	Model	Size	ET	HT	ID	IT	QU	SW	TA	TH	TR	VI	ZH	AVG
Zero-Shot	BLOOM	176B	-	-	57.5*	-	-	59.5*	54.7*	-	-	58.2*	57.7*	-
	XGLM	7.5B	57.6	57.0	59.0	49.2	52.4	55.0	55.6	57.8	55.0	59.0	53.6	55.6
	mGPT	13B	49.8	50.4	63.4	61.6	50.4	57.6	57.0	54.0	58.2	60.4	54.6	56.1
	Nemotron-4	15B	62.8	47.4	66.6	67.0	53.8	50.4	62.0	59.6	57.4	65.2	62.2	**59.5**
4-Shot	XGLM	7.5B	64.7	60.4	67.3	64.0	50.0	61.8	56.7	61.5	60.1	68.5	59.9	61.4
	mGPT	13B	48.6	48.6	62.6	60.8	50.6	56.6	55.4	54.8	57.4	61.8	58.4	56.0
	Nemotron-4	15B	72.9	52.8	79.6	79.2	50.2	52.2	72.8	66.6	77.2	78.6	76.0	**68.9**

Table 7: Comparison of Nemotron-4 15B against existing large language models on XCOPA under the zero- and four-shot setting. Our reported results for XGLM are from the runs of the model in (Shliazhko et al., 2022) given that we use the same prompt template used by mGPT. The values marked with $*$ are read from figures in (Scao et al., 2023).

2. Generation: We consider two generative tasks: TyDiQA-GoldP (Clark et al., 2020) and Multilingual Grade School Math (MGSM) (Shi et al., 2022). TyDiQA-GoldP is a question answering task while MGSM evaluates the arithmetic reasoning ability of language models in 10 languages.

In comparing the performance of Nemotron-4 15B on TyDiQA-GoldP to a range of models, Table 8 shows that Nemotron-4 15B achieves the best performance. Impressively, Nemotron-4 15B is able to significantly improve upon the next best model, PaLM 62B-cont.

Model	Size	AR	BN	FI	ID	KO	RU	SW	TE	AVG
PaLM	62B	31.2	42.5	41.7	41.6	49.3	29.2	58.1	30.6	40.5
	62B-cont	39.4	48.7	44.0	49.2	52.5	35.6	60.9	35.3	45.7
LLaMA-2	13B	-	-	-	-	-	-	-	-	33.2
Baichuan-2	13B	-	-	-	-	-	-	-	-	30.8
QWEN	14B	-	-	-	-	-	-	-	-	39.8
Nemotron-4	15B	39.1	55.8	52.2	54.5	55.1	37.8	54.5	55.0	**50.5**

Table 8: Comparative results in the one-shot setting on TyDiQA-GoldP. Results for LLaMA-2 13B, Baichuan-2 13B and QWEN 14B are taken from (Chen et al., 2024).

Further demonstrating the impressive multilingual ability of Nemotron-4 15B, Table 9 shows the performance on MGSM. We report using the English chain-of-thought setting introduced in (Shi et al., 2022) where all chain of thought explanations are presented to the model in English rather than in the language of the task. On this challenging task which assesses the intersection of mathematical and multilingual ability, Nemotron-4 15B achieves the best performance amongst compared models and improves upon the closest score by nearly 30%.

Mode	Model	Size	DE	FR	ES	RU	ZH	JA	TH	TE	BN	SW	AVG
Native-COT	PaLM	62B	24.0	24.0	26.0	22.8	24.8	14.8	18.0	11.6	13.6	9.6	18.9
English-COT	PALM	62B-cont	44.8	39.2	44.4	36.8	33.6	24.0	28.0	19.6	28.0	21.2	32.0
	Mistral	7B	33.2	35.2	35.6	35.2	33.2	18.8	10.0	0.0	8.0	9.2	21.8
	Nemotron-4	15B	46.8	46.0	50.0	45.6	40.0	40.0	43.6	41.6	43.6	16.0	**41.3**

Table 9: Eight-shot accuracy results on MGSM. Results for Mistral are from our runs of Mistral in the same setting as Nemotron-4.

3. Machine Translation: We additionally evaluate the translation ability of our models through the FLORES-101 (Goyal et al., 2021) benchmark. The ability to translate between languages is a good test of the model's ability to relate and understand semantic relationships between languages.

As seen in Table 10, Nemotron-4 15B heftily outperforms both LLaMA-2 13B and Baichuan-2 13B – improving upon their performance by 90.2% and 44.1% respectively. Nemotron-4 15B does not solely perform well on translating from Chinese into English but is able to attain impressive results on the direct translation of Chinese into other languages. This ability highlights the strong understanding that Nemotron-4 15B has across a broad spectrum of natural languages.

	Size	ZH-EN	ZH-FR	ZH-ES	ZH-AR	ZH-RU	ZH-JA	ZH-DE	AVG
LLaMA-2	13B	25.4	19.2	17.5	1.4	10.3	0.1	11.1	12.2
Baichuan-2	13B	30.6	22.1	17.3	2.4	14.2	11.6	14.5	16.1
Nemotron-4	15B	34.0	28.1	21.3	16.8	21.2	23.1	18.1	**23.2**

Table 10: Eight-shot results on Flores sub-tasks translating out of Chinese. All results for external models were obtained from (Yang et al., 2023)

4 Conclusion

We present Nemotron-4 15B, a decoder-only transformer-based large language model. It is trained on 8 trillion tokens spanning English, 53 additional natural languages as well as 43 programming languages. Nemotron-4 15B exhibits the strongest multilingual performance of any general purpose language model at its scale – even outperforming models specialized for the multilingual domain. Nemotron-4 demonstrates that pre-training sets for large language models can continue to be scaled up even further in order to improve the abilities of models.

References

NVLink and NVSwitch. `https://www.nvidia.com/en-us/data-center/nvlink/`.

Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints. *arXiv preprint arXiv:2305.13245*, 2023.

Loubna Ben Allal, Raymond Li, Denis Kocetkov, Chenghao Mou, Christopher Akiki, Carlos Munoz Ferrandis, Niklas Muennighoff, Mayank Mishra, Alex Gu, Manan Dey, Logesh Kumar Umapathi, Carolyn Jane Anderson, Yangtian Zi, Joel Lamy Poirier, Hailey Schoelkopf, Sergey Troshin, Dmitry Abulkhanov, Manuel Romero, Michael Lappert, Francesco De Toni, Bernardo García del Río, Qian Liu, Shamik Bose, Urvashi Bhattacharyya, Terry Yue Zhuo, Ian Yu, Paulo Villegas, Marco Zocca, Sourab Mangrulkar, David Lansky, Huu Nguyen, Danish Contractor, Luis Villa, Jia Li, Dzmitry Bahdanau, Yacine Jernite, Sean Hughes, Daniel Fried, Arjun Guha, Harm de Vries, and Leandro von Werra. SantaCoder: Don't Reach for the Stars!, 2023.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program Synthesis with Large Language Models, 2021.

Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. Qwen Technical Report. *arXiv preprint arXiv:2309.16609*, 2023.

Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. PIQA: Reasoning about Physical Commonsense in Natural Language. In *AAAI*, 2020.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss,

Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language Models are Few-Shot Learners. In Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html.

Federico Cassano, John Gouwar, Daniel Nguyen, Sydney Nguyen, Luna Phipps-Costin, Donald Pinckney, Ming-Ho Yee, Yangtian Zi, Carolyn Jane Anderson, Molly Q Feldman, Arjun Guha, Michael Greenberg, and Abhinav Jangda. MultiPL-E: A Scalable and Polyglot Approach to Benchmarking Neural Code Generation. *IEEE Transactions on Software Engineering*, pages 1–17, 2023a. doi: 10.1109/TSE.2023.3267446.

Federico Cassano, John Gouwar, Daniel Nguyen, Sydney Nguyen, Luna Phipps-Costin, Donald Pinckney, Ming-Ho Yee, Yangtian Zi, Carolyn Jane Anderson, Molly Q Feldman, et al. Multipl-e: a scalable and polyglot approach to benchmarking neural code generation. *IEEE Transactions on Software Engineering*, 2023b.

Du Chen, Yi Huang, Xiaopu Li, Yongqiang Li, Yongqiang Liu, Haihui Pan, Leichao Xu, Dacheng Zhang, Zhipeng Zhang, and Kun Han. Orion-14B: Open-source Multilingual Large Language Models, 2024.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating Large Language Models Trained on Code, 2021.

Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. PaLM: Scaling Language Modeling with Pathways. *arXiv preprint arXiv:2204.02311*, 2022.

Jonathan H. Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. TyDi QA: A Benchmark for Information-Seeking Question Answering in Typologically Diverse Languages. *CoRR*, abs/2003.05002, 2020. URL https://arxiv.org/abs/2003.05002.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think You have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge. *arXiv preprint arXiv:1803.05457*, 2018.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training Verifiers to Solve Math Word Problems. *CoRR*, abs/2110.14168, 2021. URL https://arxiv.org/abs/2110.14168.

Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, Jason Phang, Laria Reynolds, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A Framework for Few-shot Language Model Evaluation, September 2021. URL https://doi.org/10.5281/zenodo.5371628.

Google DeepMind Gemma Team. Gemma: Open Models Based on Gemini Research and Technology, 2024.

Google. Gemini: A Family of Highly Capable Multimodal Models, 2023.

Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc'Aurelio Ranzato, Francisco Guzmán, and Angela Fan. The FLORES-101 Evaluation Benchmark for Low-Resource and Multilingual Machine Translation. *CoRR*, abs/2106.03193, 2021. URL https://arxiv.org/abs/2106.03193.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring Massive Multitask Language Understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training Compute-Optimal Large Language Models. *arXiv preprint arXiv:2203.15556*, 2022.

Joseph Jennings, Mostofa Patwary, Sandeep Subramanian, Shrimai Prabhumoye, Ayush Dattagupta, Mohammad Shoeybi, and Bryan Catanzaro. Curating Trillion-Token Datasets: Introducing NVIDIA NeMo Data Curator. https://developer.nvidia.com/blog/curating-trillion-token-datasets-introducing-nemo-data-curator/, 2023.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling Laws for Neural Language Models. *arXiv preprint arXiv:2001.08361*, 2020.

Vijay Korthikanti, Jared Casper, Sangkug Lym, Lawrence McAfee, Michael Andersch, Mohammad Shoeybi, and Bryan Catanzaro. Reducing Activation Recomputation in Large Transformer Models, 2022.

Taku Kudo and John Richardson. Sentencepiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing. *arXiv preprint arXiv:1808.06226*, 2018.

Raymond Li, Loubna Ben Allal, Yangtian Zi, Niklas Muennighoff, Denis Kocetkov, Chenghao Mou, Marc Marone, Christopher Akiki, Jia Li, Jenny Chim, Qian Liu, Evgenii Zheltonozhskii, Terry Yue Zhuo, Thomas Wang, Olivier Dehaene, Mishig Davaadorj, Joel Lamy-Poirier, João Monteiro, Oleh Shliazhko, Nicolas Gontier, Nicholas Meade, Armel Zebaze, Ming-Ho Yee, Logesh Kumar Umapathi, Jian Zhu, Benjamin Lipkin, Muhtasham Oblokulov, Zhiruo Wang, Rudra Murthy, Jason Stillerman, Siva Sankalp Patel, Dmitry Abulkhanov, Marco Zocca, Manan Dey, Zhihan Zhang, Nour Fahmy, Urvashi Bhattacharyya, Wenhao Yu, Swayam Singh, Sasha Luccioni, Paulo Villegas, Maxim Kunakov, Fedor Zhdanov, Manuel Romero, Tony Lee, Nadav Timor, Jennifer Ding, Claire Schlesinger, Hailey Schoelkopf, Jan Ebert, Tri

Dao, Mayank Mishra, Alex Gu, Jennifer Robinson, Carolyn Jane Anderson, Brendan Dolan-Gavitt, Danish Contractor, Siva Reddy, Daniel Fried, Dzmitry Bahdanau, Yacine Jernite, Carlos Muñoz Ferrandis, Sean Hughes, Thomas Wolf, Arjun Guha, Leandro von Werra, and Harm de Vries. StarCoder: May the Source be with You!, 2023.

Xi Victoria Lin, Todor Mihaylov, Mikel Artetxe, Tianlu Wang, Shuohui Chen, Daniel Simig, Myle Ott, Naman Goyal, Shruti Bhosale, Jingfei Du, Ramakanth Pasunuru, Sam Shleifer, Punit Singh Koura, Vishrav Chaudhary, Brian O'Horo, Jeff Wang, Luke Zettlemoyer, Zornitsa Kozareva, Mona Diab, Veselin Stoyanov, and Xian Li. Few-shot Learning with Multilingual Language Models, 2022.

NVIDIA. H100 Tensor Core GPU Architecture Overview, 2022.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: A Method for Automatic Evaluation of Machine Translation. In *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics*, ACL '02, page 311–318, USA, 2002. Association for Computational Linguistics. doi: 10.3115/1073083.1073135. URL `https://doi.org/10.3115/1073083.1073135`.

Edoardo Maria Ponti, Goran Glavas, Olga Majewska, Qianchu Liu, Ivan Vulic, and Anna Korhonen. XCOPA: A Multilingual Dataset for Causal Commonsense Reasoning. *CoRR*, abs/2005.00333, 2020. URL `https://arxiv.org/abs/2005.00333`.

Matt Post. A Call for Clarity in Reporting BLEU Scores. *CoRR*, abs/1804.08771, 2018. URL `http://arxiv.org/abs/1804.08771`.

Jack W. Rae, Sebastian Borgeaud, Trevor Cai, Katie Millican, Jordan Hoffmann, Francis Song, John Aslanides, Sarah Henderson, Roman Ring, Susannah Young, Eliza Rutherford, Tom Hennigan, Jacob Menick, Albin Cassirer, Richard Powell, George van den Driessche, Lisa Anne Hendricks, Maribeth Rauh, Po-Sen Huang, Amelia Glaese, Johannes Welbl, Sumanth Dathathri, Saffron Huang, Jonathan Uesato, John Mellor, Irina Higgins, Antonia Creswell, Nat McAleese, Amy Wu, Erich Elsen, Siddhant Jayakumar, Elena Buchatskaya, David Budden, Esme Sutherland, Karen Simonyan, Michela Paganini, Laurent Sifre, Lena Martens, Xiang Lorraine Li, Adhiguna Kuncoro, Aida Nematzadeh, Elena Gribovskaya, Domenic Donato, Angeliki Lazaridou, Arthur Mensch, Jean-Baptiste Lespiau, Maria Tsimpoukelli, Nikolai Grigorev, Doug Fritz, Thibault Sottiaux, Mantas Pajarskas, Toby Pohlen, Zhitao Gong, Daniel Toyama, Cyprien de Masson d'Autume, Yujia Li, Tayfun Terzi, Vladimir Mikulik, Igor Babuschkin, Aidan Clark, Diego de Las Casas, Aurelia Guy, Chris Jones, James Bradbury, Matthew Johnson, Blake Hechtman, Laura Weidinger, Iason Gabriel, William Isaac, Ed Lockhart, Simon Osindero, Laura Rimell, Chris Dyer, Oriol Vinyals, Kareem Ayoub, Jeff Stanway, Lorrayne Bennett, Demis Hassabis, Koray Kavukcuoglu, and Geoffrey Irving. Scaling Language Models: Methods, Analysis & Insights from Training Gopher, 2022.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *The Journal of Machine Learning Research*, 21(1):5485–5551, 2020.

Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. WINOGRANDE: An Adversarial Winograd Schema Challenge at Scale. In *AAAI*, 2020.

Maarten Sap, Hannah Rashkin, Derek Chen, Ronan LeBras, and Yejin Choi. Socialiqa: Commonsense reasoning about social interactions, 2019.

Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, Matthias Gallé, Jonathan Tow, Alexander M. Rush, Stella Biderman, Albert Webson, Pawan Sasanka Ammanamanchi, Thomas Wang, Benoît Sagot, Niklas Muennighoff, Albert Villanova del Moral, Olatunji Ruwase, Rachel Bawden, Stas Bekman, Angelina McMillan-Major, Iz Beltagy, Huu Nguyen, Lucile Saulnier, Samson Tan, Pedro Ortiz Suarez, Victor Sanh, Hugo Laurençon, Yacine Jernite, Julien Launay, Margaret Mitchell, Colin Raffel, Aaron Gokaslan, Adi Simhi, Aitor Soroa, Alham Fikri Aji, Amit Alfassy, Anna Rogers, Ariel Kreisberg Nitzav, Canwen Xu, Chenghao Mou, Chris Emezue, Christopher Klamm, Colin Leong, Daniel van Strien, David Ifeoluwa Adelani, Dragomir Radev, Eduardo González Ponferrada, Efrat Levkovizh, Ethan Kim, Eyal Bar Natan, Francesco De Toni, Gérard Dupont, Germán Kruszewski, Giada Pistilli, Hady Elsahar, Hamza Benyamina, Hieu Tran, Ian Yu, Idris Abdulmumin, Isaac Johnson, Itziar Gonzalez-Dios, Javier de la Rosa, Jenny Chim, Jesse Dodge, Jian Zhu, Jonathan Chang, Jörg Frohberg, Joseph Tobing, Joydeep Bhattacharjee, Khalid Almubarak, Kimbo Chen, Kyle Lo, Leandro Von Werra, Leon Weber, Long Phan, Loubna Ben allal, Ludovic Tanguy, Manan Dey, Manuel Romero Muñoz, Maraim Masoud, María Grandury, Mario Šaško, Max Huang, Maximin Coavoux, Mayank Singh, Mike Tian-Jian Jiang, Minh Chien Vu, Mohammad A. Jauhar, Mustafa Ghaleb, Nishant Subramani, Nora Kassner, Nurulaqilla Khamis, Olivier Nguyen, Omar Espejel, Ona de Gibert, Paulo Villegas, Peter Henderson, Pierre Colombo, Priscilla Amuok, Quentin Lhoest, Rheza Harliman, Rishi Bommasani, Roberto Luis López, Rui Ribeiro, Salomey Osei, Sampo Pyysalo, Sebastian Nagel, Shamik Bose, Shamsuddeen Hassan Muhammad, Shanya Sharma, Shayne Longpre, Somaieh Nikpoor, Stanislav Silberberg, Suhas Pai, Sydney Zink, Tiago Timponi Torrent, Timo Schick, Tristan Thrush, Valentin Danchev, Vassilina Nikoulina, Veronika Laippala, Violette Lepercq, Vrinda Prabhu, Zaid Alyafeai, Zeerak Talat, Arun Raja, Benjamin Heinzerling, Chenglei Si, Davut Emre Taşar, Elizabeth Salesky, Sabrina J. Mielke, Wilson Y. Lee, Abheesht Sharma, Andrea Santilli, Antoine Chaffin, Arnaud Stiegler, Debajyoti Datta, Eliza Szczechla, Gunjan Chhablani, Han Wang, Harshit Pandey, Hendrik Strobelt, Jason Alan Fries, Jos Rozen, Leo Gao, Lintang Sutawika, M Saiful Bari, Maged S. Al-shaibani, Matteo Manica, Nihal Nayak, Ryan Teehan, Samuel Albanie, Sheng Shen, Srulik Ben-David, Stephen H. Bach, Taewoon Kim, Tali Bers, Thibault Fevry, Trishala Neeraj, Urmish Thakker, Vikas Raunak, Xiangru Tang, Zheng-Xin Yong, Zhiqing Sun, Shaked Brody, Yallow Uri, Hadar Tojarieh, Adam Roberts, Hyung Won Chung, Jaesung Tae, Jason Phang, Ofir Press, Conglong Li, Deepak Narayanan, Hatim Bourfoune, Jared Casper, Jeff Rasley, Max Ryabinin, Mayank Mishra, Minjia Zhang, Mohammad Shoeybi, Myriam Peyrounette, Nicolas Patry, Nouamane Tazi, Omar Sanseviero, Patrick von Platen, Pierre Cornette, Pierre François Lavallée, Rémi Lacroix, Samyam Rajbhandari, Sanchit Gandhi, Shaden Smith, Stéphane Requena, Suraj Patil, Tim Dettmers, Ahmed Baruwa, Amanpreet Singh, Anastasia Cheveleva, Anne-Laure Ligozat, Arjun Subramonian, Aurélie Névéol, Charles Lovering, Dan Garrette, Deepak Tunuguntla, Ehud Reiter, Ekaterina Taktasheva, Ekaterina Voloshina, Eli Bogdanov, Genta Indra Winata, Hailey Schoelkopf, Jan-Christoph Kalo, Jekaterina Novikova, Jessica Zosa Forde, Jordan Clive, Jungo Kasai, Ken Kawamura, Liam Hazan, Marine Carpuat, Miruna Clinciu, Najoung Kim, Newton Cheng, Oleg Serikov, Omer Antverg, Oskar van der Wal, Rui Zhang, Ruochen Zhang, Sebastian Gehrmann, Shachar Mirkin, Shani Pais, Tatiana Shavrina, Thomas Scialom, Tian Yun, Tomasz Limisiewicz, Verena Rieser, Vitaly Protasov, Vladislav Mikhailov, Yada Pruksachatkun, Yonatan Belinkov, Zachary Bamberger, Zdeněk Kasner, Alice Rueda, Amanda Pestana, Amir Feizpour, Ammar Khan, Amy Faranak, Ana Santos, Anthony Hevia, Antigona Unldreaj, Arash Aghagol, Arezoo Abdollahi, Aycha Tammour, Azadeh HajiHosseini, Bahareh Behroozi, Benjamin Ajibade, Bharat Saxena, Carlos Muñoz Ferrandis, Danish Contractor, David Lansky, Davis David, Douwe Kiela, Duong A. Nguyen, Edward Tan, Emi Baylor, Ezinwanne Ozoani, Fatima Mirza, Frankline Ononiwu, Habib Rezane-

jad, Hessie Jones, Indrani Bhattacharya, Irene Solaiman, Irina Sedenko, Isar Nejadgholi, Jesse Passmore, Josh Seltzer, Julio Bonis Sanz, Livia Dutra, Mairon Samagaio, Maraim Elbadri, Margot Mieskes, Marissa Gerchick, Martha Akinlolu, Michael McKenna, Mike Qiu, Muhammed Ghauri, Mykola Burynok, Nafis Abrar, Nazneen Rajani, Nour Elkott, Nour Fahmy, Olanrewaju Samuel, Ran An, Rasmus Kromann, Ryan Hao, Samira Alizadeh, Sarmad Shubber, Silas Wang, Sourav Roy, Sylvain Viguier, Thanh Le, Tobi Oyebade, Trieu Le, Yoyo Yang, Zach Nguyen, Abhinav Ramesh Kashyap, Alfredo Palasciano, Alison Callahan, Anima Shukla, Antonio Miranda-Escalada, Ayush Singh, Benjamin Beilharz, Bo Wang, Caio Brito, Chenxi Zhou, Chirag Jain, Chuxin Xu, Clémentine Fourrier, Daniel León Periñán, Daniel Molano, Dian Yu, Enrique Manjavacas, Fabio Barth, Florian Fuhrimann, Gabriel Altay, Giyaseddin Bayrak, Gully Burns, Helena U. Vrabec, Imane Bello, Ishani Dash, Jihyun Kang, John Giorgi, Jonas Golde, Jose David Posada, Karthik Rangasai Sivaraman, Lokesh Bulchandani, Lu Liu, Luisa Shinzato, Madeleine Hahn de Bykhovetz, Maiko Takeuchi, Marc Pàmies, Maria A Castillo, Marianna Nezhurina, Mario Sänger, Matthias Samwald, Michael Cullan, Michael Weinberg, Michiel De Wolf, Mina Mihaljcic, Minna Liu, Moritz Freidank, Myungsun Kang, Natasha Seelam, Nathan Dahlberg, Nicholas Michio Broad, Nikolaus Muellner, Pascale Fung, Patrick Haller, Ramya Chandrasekhar, Renata Eisenberg, Robert Martin, Rodrigo Canalli, Rosaline Su, Ruisi Su, Samuel Cahyawijaya, Samuele Garda, Shlok S Deshmukh, Shubhanshu Mishra, Sid Kiblawi, Simon Ott, Sinee Sang-aroonsiri, Srishti Kumar, Stefan Schweter, Sushil Bharati, Tanmay Laud, Théo Gigant, Tomoya Kainuma, Wojciech Kusa, Yanis Labrak, Yash Shailesh Bajaj, Yash Venkatraman, Yifan Xu, Yingxin Xu, Yu Xu, Zhe Tan, Zhongli Xie, Zifan Ye, Mathilde Bras, Younes Belkada, and Thomas Wolf. BLOOM: A 176B-Parameter Open-Access Multilingual Language Model, 2023.

Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. Language Models are Multilingual Chain-of-Thought Reasoners, 2022.

Oleh Shliazhko, Alena Fenogenova, Maria Tikhonova, Vladislav Mikhailov, Anastasia Kozlova, and Tatiana Shavrina. mGPT: Few-Shot Learners Go Multilingual, 2022.

Mohammad Shoeybi, Mostofa Patwary, Raul Puri, Patrick LeGresley, Jared Casper, and Bryan Catanzaro. Megatron-LM: Training Multi-Billion Parameter Language Models using Model Parallelism. *arXiv preprint arXiv:1909.08053*, 2019.

Andrew M. Dai Slav Petrov, Yonghui Wu and et al. PaLM 2 Technical Report, 2023. URL `https://ai.google/static/documents/palm2techreport.pdf`.

Shaden Smith, Mostofa Patwary, Brandon Norick, Patrick LeGresley, Samyam Rajbhandari, Jared Casper, Zhun Liu, Shrimai Prabhumoye, George Zerveas, Vijay Korthikanti, Elton Zheng, Rewon Child, Reza Yazdani Aminabadi, Julie Bernauer, Xia Song, Mohammad Shoeybi, Yuxiong He, Michael Houston, Saurabh Tiwary, and Bryan Catanzaro. Using DeepSpeed and Megatron to Train Megatron-Turing NLG 530B, A Large-Scale Generative Language Model. *CoRR*, abs/2201.11990, 2022. URL `https://arxiv.org/abs/2201.11990`.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced Transformer with Rotary Position Embedding. *arXiv preprint arXiv:2104.09864*, 2021.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. Challenging big-bench tasks and whether chain-of-thought can solve them, 2022.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. LLaMA: Open and Efficient Foundation Language Models, 2023a.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open Foundation and Fine-tuned Chat Models. *arXiv preprint arXiv:2307.09288*, 2023b.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. *CoRR*, abs/1706.03762, 2017. URL `http://arxiv.org/abs/1706.03762`.

Guillaume Wenzek, Marie-Anne Lachaux, Alexis Conneau, Vishrav Chaudhary, Francisco Guzmán, Armand Joulin, and Edouard Grave. CCNet: Extracting High Quality Monolingual Datasets from Web Crawl Data. *arXiv preprint arXiv:1911.00359*, 2019.

Aiyuan Yang, Bin Xiao, Bingning Wang, Borong Zhang, Chao Yin, Chenxu Lv, Da Pan, Dian Wang, Dong Yan, Fan Yang, et al. Baichuan 2: Open Large-scale Language Models. *arXiv preprint arXiv:2309.10305*, 2023.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. HellaSwag: Can a Machine Really Finish Your Sentence? In *ACL*, 2019.

Supplementary Materials

	Size	Humanities	Social sciences	STEM	Other	Average
Nemotron-4	15B	69.2	74.1	53.4	67.5	64.2

Table 11: Per-category breakdown accuracy for MMLU